

Computer Organization and Architecture (24B41EC312)



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Module 1:

STRUCTURE OF COMPUTERS:

- Generation of Computers,
- Computer types,
- Classification of Computers, functional units, basic operational concepts,
- Von-Neumann architecture, bus structures, types of bus, software,
- Multiprocessors and multicomputer.

What is Computer Architecture & Computer Organization

- **Computer Architecture:**

Computer Architecture deals with giving operational attributes of the computer or Processor to be specific. It deals with details like physical memory, ISA (Instruction Set Architecture) of the processor, the number of bits used to represent the data types, Input Output mechanism and technique for addressing memories.

- Computer Organization:**

Computer Organization is realization of what is specified by the computer architecture .It deals with how operational attributes are linked together to meet the requirements specified by computer architecture. Some organizational attributes are hardware details, control signals, peripherals.

Difference between Computer Architecture and Computer Organization

Computer Architecture	Computer Organization
Architecture describes what the computer does.	The Organization describes how it does it.
Computer Architecture deals with the functional behavior of computer systems.	Computer Organization deals with a structural relationship.
In the above figure, it's clear that it deals with high-level design issues.	In the above figure, it's also clear that it deals with low-level design issues.
Architecture indicates its hardware.	Whereas Organization indicates its performance.
As a programmer, you can view architecture as a series of instructions, addressing modes, and registers.	The implementation of the architecture is called organization.
For designing a computer, its architecture is fixed first.	For designing a computer, an organization is decided after its architecture.

Computer Architecture comprises logical functions such as instruction sets, registers, data types, and addressing modes .	Computer Organization consists of physical units like circuit designs, peripherals, and adders.
The different architectural categories found in our computer systems are as follows: • Von-Neumann Architecture • Harvard Architecture • Instruction Set Architecture • Micro-architecture • System Design	CPU organization is classified into three categories based on the number of address fields: • Organization of a single Accumulator. • Organization of general registers • Stack organization
It makes the computer's hardware visible.	It offers details on how well the computer performs.
Architecture coordinates the hardware and software of the system.	Computer Organization handles the segments of the network in a system.
The software developer is aware of it.	It escapes the software programmer's detection.
Examples- Intel and AMD created the x86 processor. Sun Microsystems and others created the SPARC processor. Apple, IBM, and Motorola created the PowerPC.	Organizational qualities include hardware elements that are invisible to the programmer, such as interfacing of computer and peripherals, memory technologies, and control signals.

Generation of Computers

- Generation in computer terminology is a change in technology a computer is/was being used. Initially, the generation term was used to distinguish between varying hardware technologies. But nowadays, generation includes both hardware and software, which together make up an entire computer system.

- There are totally five computer generations known till date. Each generation has been discussed in detail along with their time period and characteristics

- Following are the main five generations of computers

S.N.	Generation & Description
1	First Generation The period of first generation: 1946-1959. Vacuum tube based.
2	Second Generation The period of second generation: 1959-1965. Transistor based.
3	Third Generation The period of third generation: 1965-1971. Integrated Circuit based.
4	Fourth Generation The period of fourth generation: 1971-1980. VLSI microprocessor based.
5	Fifth Generation The period of fifth generation: 1980-onwards. ULSI microprocessor based

First generation

- The period of first generation was 1946-1959.
- The computers of first generation used vacuum tubes as the basic components for memory and circuitry for CPU (Central Processing Unit). These tubes, like electric bulbs, produced a lot of heat and were prone to frequent fusing of the installations, therefore, were very expensive and could be afforded only by very large organizations.

- In this generation mainly batch processing operating system were used. Punched cards, paper tape, and magnetic tape were used as input and output devices.
- The computers in this generation used machine code as programming language.

The main features of first generation are:

- Vacuum tube technology
- Unreliable
- Supported machine language only
- Very costly
- Generated lot of heat
- Slow input and output devices
- Huge size
- Need of A.C.
- Non-portable
- Consumed lot of electricity



- **Some computers of this generation were:**

- . ENIAC
- . EDVAC
- . UNIVAC
- . IBM-701
- . IBM-650

Second generation

- The period of second generation was 1959-1965. In this generation transistors were used that were cheaper, consumed less power, more compact in size, more reliable and faster than the first-generation machines made of vacuum tubes.
- In this generation, magnetic cores were used as primary memory and magnetic tape and magnetic disks as secondary storage devices.

- In this generation assembly language and high-level programming languages like FORTRAN, COBOL were used.
- The computers used batch processing and multiprogramming operating system

The main features of second generation are:

- Use of transistors
- Reliable in comparison to first generation computers
- Smaller size as compared to first generation computers
- Generated less heat as compared to first generation computers
- Consumed less electricity as compared to first generation computers
- Faster than first generation computers
- Still very costly
- A.C. needed
- Supported machine and assembly languages



Some computers of this generation were:

- IBM 1620
- IBM 7094
- CDC 1604
- CDC 3600
- UNIVAC 1108

Third generation

- The period of third generation was 1965-1971. The computers of third generation used integrated circuits (IC's) in place of transistors.
- A single IC has many transistors, resistors and capacitors along with the associated circuitry. The IC was invented by Jack Kilby. This development made computers smaller in size, reliable and efficient.
- In this generation remote processing, time-sharing, multi-programming operating system were used. High-level languages (FORTRAN-II TO IV, COBOL, PASCAL PL/1, BASIC, ALGOL-68 etc.) were used during this generation.

The main features of third generation are:

- . IC used
- . More reliable in comparison to previous two generations
- . Smaller size
- . Generated less heat
- . Faster
- . Lesser maintenance
- . Still costly
- . A.C needed
- . Consumed lesser electricity



- . **Some computers of this generation were:**
 - . IBM-360 series
 - . Honeywell-6000 series
 - . PDP(Personal Data Processor)
 - . IBM-370/168
 - . TDC-316

Fourth generation

- The period of fourth generation was 1971-1980.
- The computers of fourth generation used Very Large Scale Integrated (VLSI) circuits. VLSI circuits having about 5000 transistors and other circuit elements and their associated circuits on a single chip made it possible to have microcomputers of fourth generation.

- Fourth generation computers became more powerful, compact, reliable, and affordable. As a result, it gave rise to personal computer (PC) revolution.
- In this generation time sharing, real time, networks, distributed operating system were used. All the high-level languages like C, C++, DBASE etc., were used in this generation.

The main features of fourth generation are:

- VLSI technology used
- Very cheap
- Portable and reliable
- Use of PC's
- Very small size
- Pipeline processing
- No A.C. needed
- Concept of internet was introduced
- Great developments in the fields of networks



Computers became easily available Some computers of this generation were:

- DEC 10
- STAR 1000
- PDP 11
- CRAY-1(Super Computer)
- CRAY-X-MP (Super Computer)

Fifth generation

- The period of fifth generation is 1980-till date. In the fifth generation, the VLSI technology became ULSI (Ultra Large-Scale Integration) technology, resulting in the production of microprocessor chips having ten million electronic components.
- This generation is based on parallel processing hardware and AI (Artificial Intelligence) software.
- AI is an emerging branch in computer science, which interprets means and method of making computers think like human beings. All the high-level languages like C and C++, Java, .NET etc., are used in this generation.

- AI includes:

1. Robotics
2. Neural Networks
3. Game Playing
4. Development of expert systems to make decisions in real life situations.
5. Natural language understanding and generation.

The main features of fifth generation are:

1. ULSI technology
2. Development of true artificial intelligence
3. Development of Natural language processing
4. Advancement in Parallel Processing
5. Advancement in Superconductor technology
6. More user friendly interfaces with multimedia features
7. Availability of very powerful and compact computers at cheaper rates



Some computer types of this generation are:

1. Desktop
1. Laptop
2. NoteBook
3. UltraBook
4. Chromebook

Computer types:

Classification based on Operating Principles

Based on the operating principles, computers can be classified into one of the following types:

- Digital Computers
- Analog Computers
- Hybrid Computers

- **Digital computers** use discrete binary (0s and 1s) for high accuracy, processing data in steps,
- **while analog computers** use continuous physical quantities (like voltage/pressure) for real-time speed but less precision;
- **hybrid computers** blend both, offering analog speed with digital accuracy for complex tasks like medical monitoring or flight simulators, combining continuous measurement with precise calculation.

Digital Computers: -

- Data:** Discrete, binary (on/off, 0/1).
- Processing:** Sequential, precise, high accuracy.
- Memory:** Large capacity.
- Examples:** Laptops, smartphones, desktops.
- Best For:** General computing, data storage, precise calculations.



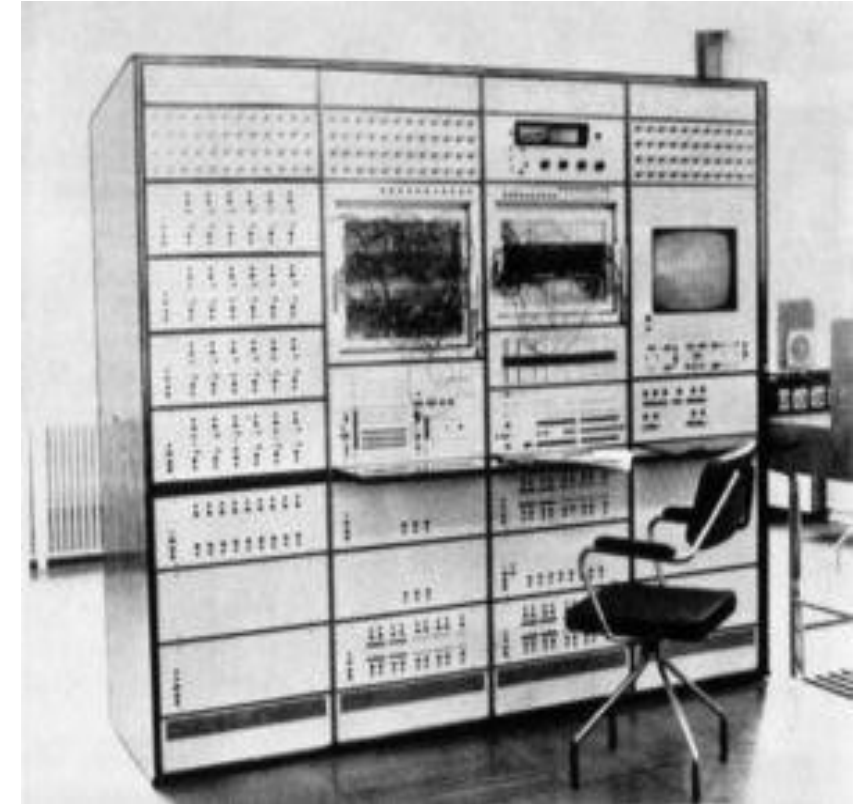
Analog Computers: -

- Data:** Continuous physical quantities (voltage, temperature, speed).
- Processing:** Real-time, parallel, less precise.
- Memory:** Limited or none (solves in one operation).
- Examples:** Speedometers, thermometers, tide predictors.
- Best For:** Scientific modeling, measuring physical phenomena.



Hybrid Computers: -

- Data:** Both continuous (analog) and discrete (digital).
- Processing:** Combines analog speed with digital precision.
- Memory:** Combines features of both.
- Examples:** Flight simulators, industrial control systems, medical monitors (ECGs).
- Best For:** Applications needing both speed and accuracy, like real-time simulations.



Classification digital Computer based on size and Capability

Based on size and capability, computers are broadly classified into

Microcomputers (PCs) are personal, small-scale devices;

Minicomputers are mid-range, multi-user systems for businesses;

Mainframes are large, powerful, high-capacity machines for big organizations; and

Supercomputers are the most powerful, fastest machines for complex scientific simulations, with differences primarily in size, processing power, cost, and intended use, forming a hierarchy from personal to massive computational needs

Microcomputers (Personal Computers)

- These are the most common systems designed for individual use.
- Purpose: Personal tasks like web browsing, word processing, and entertainment.
- Users: Supports a single user at a time.
- Hardware: Built around a single microprocessor.
- Examples: Laptops, desktop PCs, tablets, and smartphones.

Minicomputers (Midrange Computers)

- Minicomputers are mid-sized systems that act as an intermediate between personal computers and large mainframes.
- Purpose: Managing tasks for small to medium-sized businesses, such as payroll, inventory, or departmental operations.
- Users: Supports multiple users simultaneously (typically 4 to 200+) through time-sharing technology.
- Hardware: More powerful than microcomputers, often used as servers.
- Examples: IBM AS/400e, VAX series

Mainframe Computers

- These are large, high-performance systems designed for massive data processing.
- Purpose: Handling high-volume, mission-critical transactions for large organizations like banks and government agencies.
- Users: Supports thousands of users concurrently.
- Key Strength: Reliability and massive Input/Output (I/O) throughput rather than raw calculation speed.
- Examples: IBM zSeries

Role of Mainframe in ATM:

- An **Automated Teller Machine (ATM)** is not a **mainframe computer itself**, but rather a client or terminal device that connects to a bank's central **mainframe computer** system for processing transactions. The mainframe serves as the powerful, secure, and reliable backend infrastructure that handles the massive volume of transactions from thousands of ATMs and other online services.
- If you ever used an automated teller machine (ATM) to interact with your bank account, you used a mainframe. Today, mainframe computers play a central role in the daily operations of most of the world's largest corporations

- When a user interacts with an ATM, the following process, managed by the central mainframe, occurs:
- **Authentication and Request:** The ATM, which is essentially a specialized computer running its own operating system (like Windows 10 in some cases), reads the user's card and accepts their PIN and transaction request (e.g., withdrawal amount).
- **Secure Communication:** The ATM transmits this information over a secure network (often a Wide Area Network, or WAN) to a host processor, which acts as a gateway to the bank's system.

- **Mainframe Processing:** The host processor directs the request to the bank's central mainframe computer. The mainframe then:
- Verifies the user's identity and PIN against its vast database of customer accounts.
- Checks the account balance to ensure sufficient funds are available.
- Processes the transaction by updating the account records in real-time.
- Sends an approval or denial message back through the network to the specific ATM.

- Physical Dispensing/Receipt:** If approved, the ATM's internal components (cash dispenser, printer) perform the physical actions of giving out cash and a receipt, based on the instructions from the mainframe.

Classification based on number of microprocessors

- Based on the number of microprocessors, computers can be classified into
 - Sequential computers and
 - Parallel computers
- **Sequential computers:** - Sequential Computing is the type of computing where one instruction is given at a particular time and the next instruction has to wait for the first instruction to execute. It is also known as a traditional computing method because all the instructions are executed in a sequence. It has a single processor with low performance and high workload of the processor. The main disadvantage of using this computing is that it takes more time as a single instruction is getting executed at a given point in time.

- **Parallel computers:** - To remove the disadvantage of sequential computing, [parallel computing](#) was introduced. It is a type of computing where many calculations or the execution of the processes are carried out parallelly or simultaneously. The type of computing where processes can execute simultaneously at a time is known as parallel computing. It saves time as the processes are executed simultaneously. It solves larger problems. There are multiple processors with high performance and low workload per processor.

Difference Between Sequential Computing and Parallel Computing

Parameters	Sequential Computing	Parallel Computing
1.	All the instructions are executed in a sequence, one at a time.	All the instructions are executed parallelly.
2.	It has a single processor.	It is having multiple processors.
3.	It has low performance and the workload of the processor is high due to the single processor.	It has high performance and the workload of the processor is low because multiple processors are working simultaneously.
4.	Bit-by-bit format is used for data transfer.	Data transfers are in bytes.
5.	It requires more time to complete the whole process.	It requires less time to complete the whole process.
6.	Cost is low	Cost is high

Classification based on number of users

- Single-user systems serve one person at a time (like a home PC),
- multi-user systems allow several people concurrently on one machine (like a server), and
- network systems connect multiple computers to share resources (like a company LAN).

Single-User System

- Definition: Designed for one person to use at a time, dedicating all resources to that individual.
- Examples: Personal computers (Windows, macOS), smartphones, DOS.
- Characteristics: Personalized settings, dedicated performance, limited scalability for multiple users.

Multi-User System

- Definition: Allows multiple users to access and work on the same system simultaneously.
- Examples: Linux, UNIX, Windows Server, shared enterprise systems, cloud platforms.
- Characteristics: User authentication, resource management (CPU, RAM), separate profiles, enhanced security, efficient resource sharing.

Network System

- Definition: Focuses on connecting different computers (workstations) to share resources like files, printers, and internet access.
- Examples: Novell NetWare, Windows NT (as Network OS), local area networks (LANs).
- Characteristics: Establishes communication between devices, facilitates file/printer sharing, often uses client-server or peer-to-peer models.

COMPUTER TYPES:

- A computer can be defined as a fast electronic calculating machine that accepts the (data) digitized input information process it as per the list of internally stored instructions and produces the resulting information. List of instructions are called programs & internal storage is called computer memory.
- The different types of computers are
 - **Personal computers:** - This is the most common type found in homes, schools, Business offices etc., It is the most common type of desk top computers with processing and storage units along with various input and output devices.
 - **Note book computers:** - These are compact and portable versions of PC

- **Work stations:** - These have high resolution input/output (I/O) graphics capability, but with same dimensions as that of desktop computer. These are used in engineering applications of interactive design work.
- **Enterprise systems:** - These are used for business data processing in medium to large corporations that require much more computing power and storage capacity than work stations. Internet associated with servers have become a dominant worldwide source of all types of information.
- **Super computers:** - These are used for large scale numerical calculations required in the applications like weather forecasting etc.,

BASIC TERMINOLOGY

- **Input:** Whatever is put into a computer system.
- **Data:** Refers to the symbols that represent facts, objects, or ideas.
- **Information:** The results of the computer storing data as bits and bytes; the words, numbers, sounds, and graphics.
- **Output:** Consists of the processing results produced by a computer.
- **Processing:** Manipulation of the data in many ways.
- **Memory:** Area of the computer that temporarily holds data waiting to be processed, stored, or output.
- **Storage:** Area of the computer that holds data on a permanent basis when it is not immediately needed for processing.

- •**Assembly language program (ALP)** –Programs are written using mnemonics
- •**Mnemonic** –Instruction will be in the form of English like form
- •**Assembler** –is a software which converts ALP to MLL (Machine Level Language)
- •**HLL (High Level Language)** –Programs are written using English like statements
- •**Compiler** -Convert HLL to MLL, does this job by reading source program at once
- •**Interpreter** –Converts HLL to MLL, does this job statement by statement
- •**System software** –Program routines which aid the user in the execution of programs eg: Assemblers, Compilers
- •**Operating system** –Collection of routines responsible for controlling and coordinating all the activities in a computer system

Comparison of Language Types

Feature 	Machine Language	Assembly Language	High-Level Language
Abstraction Level	Lowest (no abstraction)	Low	High (strong abstraction from hardware)
Format	Binary code (0s and 1s)	Mnemonics (e.g., ADD , MOV , SUB)	English-like statements and syntax
Readability (Human)	Very difficult/nearly impossible	Difficult, but manageable with effort	Very easy/user-friendly
Execution	Directly by the CPU	Requires an assembler to translate to machine code	Requires a compiler or interpreter to translate to machine code

Portability	Not portable (hardware-specific)	Not portable (architecture-specific)	Highly portable (can run on different hardware with recompilation/interpretation)
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Execution Speed	Very fast (no translation needed at runtime)	Fast, but slower than machine code due to assembly step	Slower than assembly/machine code due to compilation/interpretation overhead
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Examples	Raw binary instruction codes	<u>x86</u> or ARM assembly	Python, C, Java, Ruby, etc.
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Computers has two kinds of components:

- **Hardware**, consisting of its physical devices (CPU, memory, bus, storage devices, ...)
- **Software**, consisting of the programs it has (Operating system, applications, utilities, ...)

FUNCTIONAL UNIT

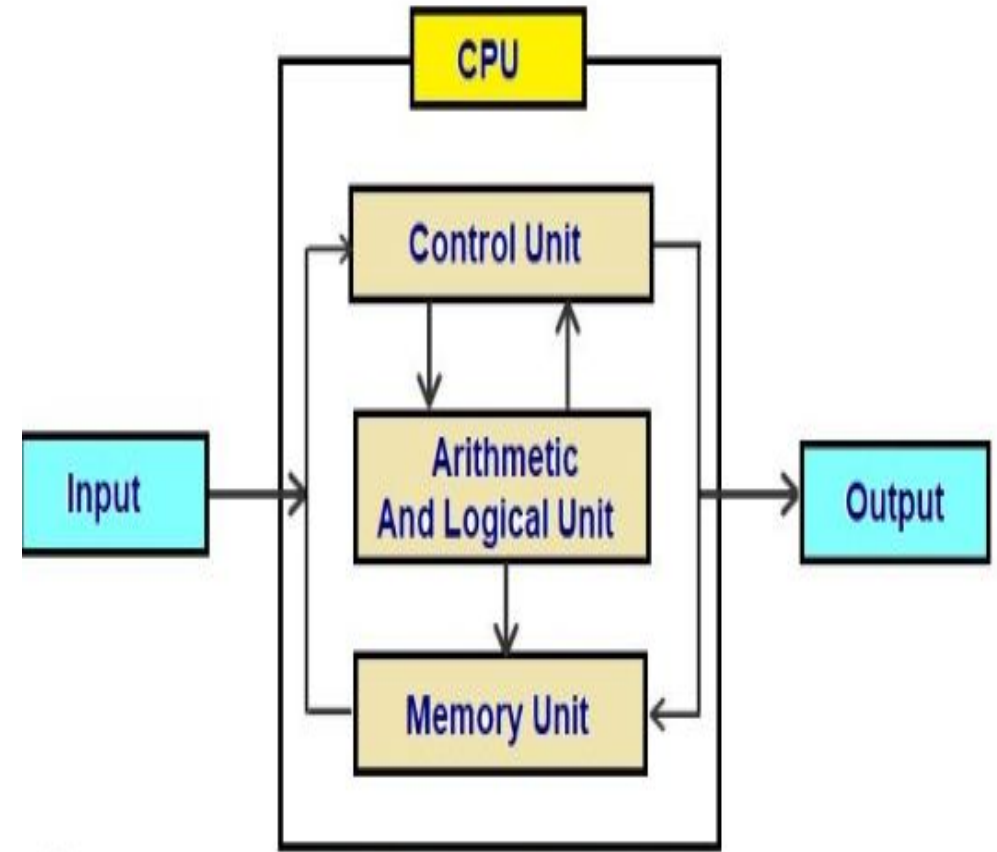
- A computer's functional units are its five core components:
- the **Input Unit** (receives data),
- **Memory Unit** (stores data/programs),
- **Arithmetic Logic Unit (ALU)** (performs calculations),
- **Control Unit (CU)** (manages operations), and
- **Output Unit** (displays results), all working together to process information under the coordination of the CU, often within the CPU, using buses for communication.

How They Work Together:

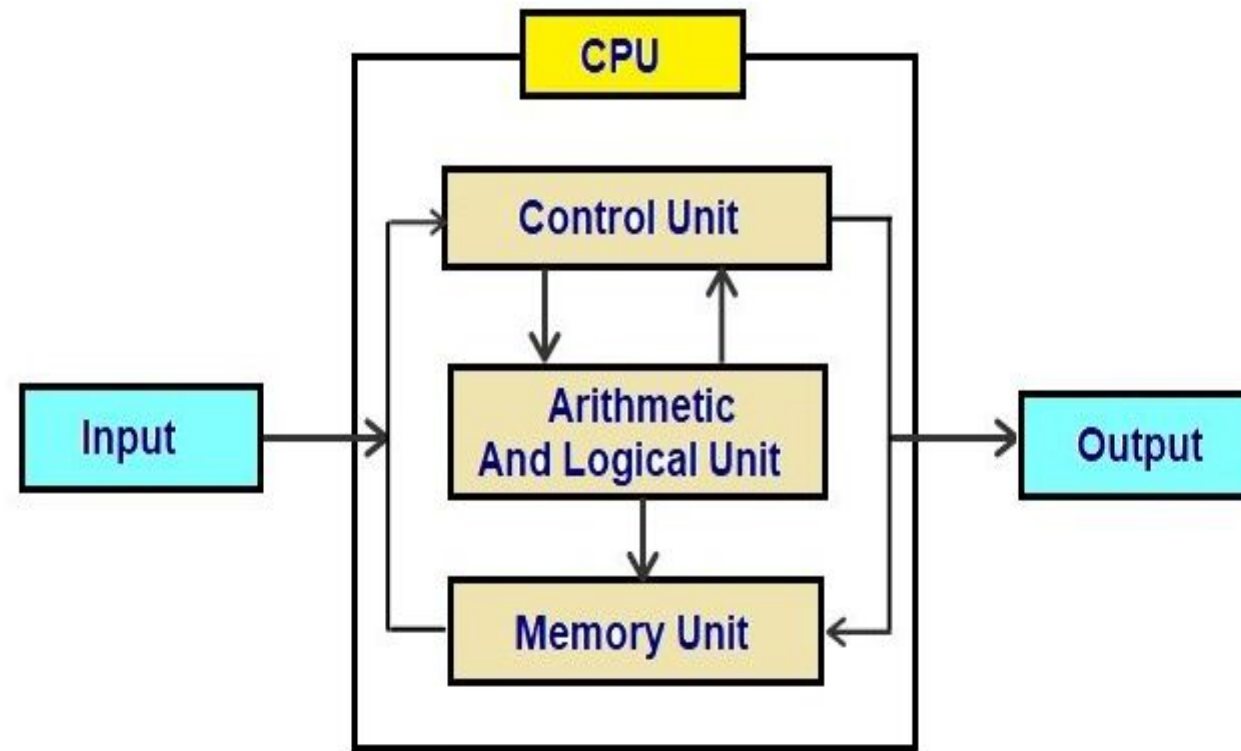
The **Input Unit** gets data into the system, which goes to **Memory**.

The **Control Unit** fetches instructions and data from **Memory**, directs the **ALU** to process them, and sends results to the Output Unit or back to Memory.

All units are interconnected by **Buses** (data, address, control) for data transfer.



Block Diagram of Computer:



Block Diagram of a Computer

Input

- All the data received by the computer goes through the [input unit](#). The input unit comprises different devices like a mouse, keyboard, scanner, etc. In other words, each of these devices acts as a mediator between the users and the computer.
- The data that is to be processed is put through the input unit. The computer accepts the raw data in binary form. It then processes the data and produces the desired output.

CPU – Central Processing Unit

- Central Processing Unit or the [CPU](#), is the brain of the computer. It works the same way a human brain works. As the brain controls all human activities, similarly the CPU controls all the tasks.
- Moreover, the CPU conducts all the arithmetical and logical operations in the computer.
- Now the CPU comprises of two units, namely – ALU (Arithmetic Logic Unit) and CU (Control Unit). Both of these units work in sync. The CPU processes the data as a whole.
- Let us see what particular tasks are assigned to both units.

ALU – Arithmetic Logic Unit

- The Arithmetic Logic Unit is made of two terms, arithmetic and logic. There are two primary functions that this unit performs.
- Data is inserted through the input unit into the primary memory. Performs the basic arithmetical operations on it, like addition, subtraction, multiplication, and division. It performs all sorts of calculations required on the data. Then, it sends back data to the storage.
- The unit is also responsible for performing logical operations like AND, OR, Equal to, Less than, etc. In addition to this, it conducts merging, sorting, and selection of the given data.

CU – Control Unit

- The control unit as the name suggests is the controller of all the activities/tasks and operations. All this is performed inside the computer.
- The [memory unit](#) sends a set of instructions to the control unit. Then the control unit in turn converts those instructions. After that these instructions are converted to control signals.

These control signals help in prioritizing and scheduling activities. Thus, the control unit coordinates the tasks inside the computer in sync with the input and output units.

Memory Unit

- All the data that has to be processed or has been processed is stored in the memory unit. The memory unit acts as a hub of all the data. It transmits it to the required part of the computer whenever necessary.
- The memory unit works in sync with the CPU. This helps in faster accessing and processing of the data. Thus, making tasks easier and quicker.

Types of Computer Memory

There are three types of computer memory-

1. **Primary Memory (Main Memory)**

Primary memory is directly accessible by the CPU and is essential for a computer to function.

RAM (Random Access Memory): A volatile memory that stores data currently being used by the CPU. It is lost when power is turned off.

DRAM (Dynamic RAM): The most common type of system memory. Modern versions include DDR4 and DDR5.

SRAM (Static RAM): Faster and more expensive than DRAM; it is primarily used for Cache memory because it doesn't need constant refreshing.

ROM (Read-Only Memory): A non-volatile memory used to store permanent instructions like the BIOS/UEFI needed to boot the computer.

PROM (Programmable ROM): Can be programmed once by the user.

EPROM (Erasable Programmable ROM): Can be erased using ultraviolet light and then reprogrammed.

EEPROM (Electrically Erasable Programmable ROM): Can be erased and rewritten electrically; modern Flash memory is a type of EEPROM.

2. Secondary Memory (Storage)

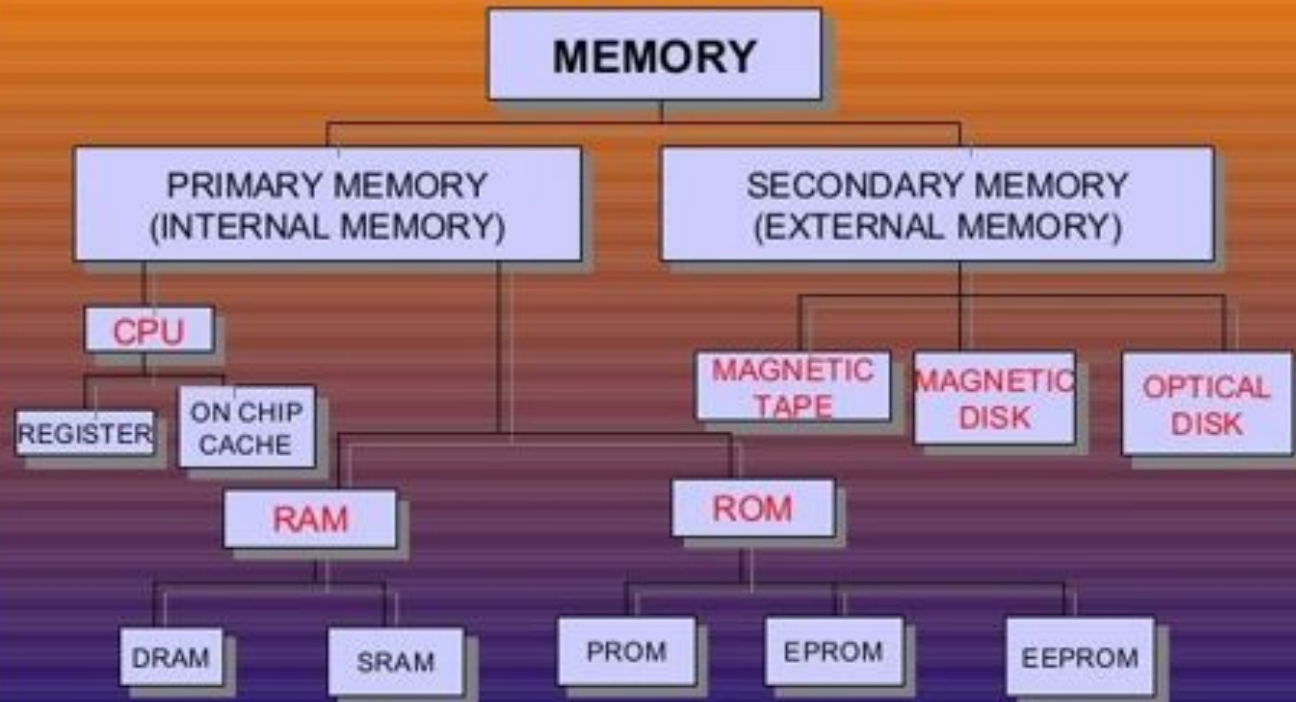
Secondary memory is non-volatile and used for long-term data storage. The CPU does not access it directly; data must first be moved to RAM.

- **HDD (Hard Disk Drive):** Uses magnetic platters to store large amounts of data at a low cost.
- **SSD (Solid State Drive):** Uses flash memory for much faster read/write speeds than HDDs and has no moving parts.
- **Optical Storage:** Includes CDs, DVDs, and Blu-ray discs that use lasers to read/write data.
- **Removable Storage:** Examples include USB flash drives and SD cards.

3. Specialty Memory

- **Cache Memory:** Extremely fast, small memory located on or very near the CPU chip. It stores frequently used data to speed up processing.
- **Registers:** The smallest and fastest storage units, located directly inside the CPU to hold immediate instructions and calculation results.
- **Virtual Memory:** A memory management technique where the operating system uses a portion of the hard drive or SSD to act as "extra" RAM when physical RAM is full.

Types of memory



Output

- There is nothing to be amazed by what the output unit is used for. All the information sent to the computer once processed is received by the user through the output unit. Devices like printers, monitors, projectors, etc. all come under the output unit.
- The output unit displays the data either in the form of a soft copy or a hard copy. The printer is for the hard copy. The monitor is for the display. The output unit accepts the data in binary form from the computer. It then converts it into a readable form for the user